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# KNOWLEDGE-BASED EXPERT SYSTEMS IN STRUCTURAL DESIGN

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D. Sriram, M.L. Maher and S.J. Fenves

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# 基于知识的结构设计专家系统

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## ABSTRACT

A targe number of problems encountered by structural designers are ill-structured and are either not amenable to an algorithmic solution or the algorithms are too cumbersome and restrictive. Knowledge-based expert systems (KBES) provide a means to assist in the solution of these ill-structured problems that demand considerable expertise. KBES also provide a flexible methodology for developing engineering software, as welt as a mechanism for formalizing thought processes about design. In this paper applications of KBES to the preliminary design, analysts, and detailed design phases of structural design are discussed. A conceptual view of an integrated structural design system in the tight of recent advances in KBES development is also presented.

To be presented at the NASA Symposium on Advances and Trends
in Structures and Dynamics, October 1984, at Washington, 0. C.

# 基于知识的结构设计专家系统

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## 摘要

结构设计师所面临的诸多问题往往属于非结构化问题，既难以通过算法求解，即便存在算法也往往过于繁琐且约束性强。基于知识的专家系统（KBES）为解决这些需要深厚专业知识的非结构化问题提供了有效途径。此外，KBES还为工程软件的开发提供了一种灵活的方法学，并为设计思维过程的规范化提供了机制。本文探讨了KBES在结构设计的初步设计、分析及详细设计阶段的应用，并结合KBES发展的最新进展，提出了一个集成化结构设计系统的概念框架。

**To be presented at the NASA Symposium on Advances and Trends in Structures and Dynamics, October 1984, at Washington, 0. C.**

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# KNOWLEDGE-BASED EXPERT SYSTEMS IN STRUCTURAL DESIGN

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## INTRODUCTION

The use of computers in structural design to date has been extensive, but limited to algorithmic analysis, proportioning and graphical presentation of results (see [8] for a survey of current CAD trends and applications in structural engineering). However, a number of problems encountered in structural design are not amenable to purely algorithmic solutions. These problems are often ill-structured, and an experienced engineer deals with them using his judgment and experience. Knowledge-based expert systems (KBES) provide a means for using the computer to assist in the solution of these ill-structured problem **areas.**

In this paper we discuss some applications of expert systems in structural design. A brief overview of KBES is provided in the next section. The design process is then briefly presented, followed by three application examples. Finally, a conceptual view of an integrated structural design and analysis system is presented incorporating recent developments in artificial intelligence and expert systems methodology.

## OVERVIEW OF KBES

Artificial Intelligence (Al) is the area of computer science concerned with the emulation of human thought processes. Efforts in the application of Al methods to intelligent problem solving led to the development of *expert systems,* systems which perform tasks that require a great deal of specialized

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# 基于知识的结构设计专家系统

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## 引言

迄今为止，计算机在结构设计中的应用已十分广泛，但主要局限于算法化分析、构件尺寸确定以及结果的图形化展示（关于当前结构工程领域CAD技术的发展趋势与应用，请参见文献[8]）。然而，结构设计中所遇到的诸多问题并不适合采用纯算法化的解决方案。这类问题往往属于非结构化问题，经验丰富的工程师通常凭借其判断力和实践经验加以处理。基于知识的专家系统（KBES）为利用计算机辅助解决这些非结构化问题提供了有效途径。

本文探讨了专家系统在结构设计中的若干应用。下一节将对基于知识的专家系统（KBES）进行简要概述。随后，我们将简要介绍设计流程，并给出三个应用实例。最后，结合人工智能与专家系统方法学的最新进展，提出一种集成化结构设计与分析系统的概念性框架。

## KBES 概述

人工智能（AI）是计算机科学的一个分支，致力于模拟人类的思维过程。将人工智能方法应用于智能问题求解的研究，促成了专家系统的开发，这类系统能够执行需要大量专业知识的任务。

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knowledge that experts in a particular field acquire from long experience with such tasks. Since these systems use domain-specific knowledge they are referred to as *knowledge-based expert systems.* Gaschnig et at. [12] define KBES as "Knowledge-based expert systems are interactive computer programs incorporating judgment experience, rules of thumb, intuition, and other expertise to provide knowledgeable advice about a variety of tasks".

The spectrum of problems encountered in the development of KBES is bounded by *derivation* and *formation* tasks [2]. In derivation problems, the problem conditions are presented as a part of the solution. These problems are normally solved by completing the solution path using available knowledge and data. Some problems encountered at this end of the spectrum are: *interpretation, diagnosis, monitoring,* and *prediction.* Formation problems are posed by the conditions or constraints that the solution as a whole must satisfy. One common way of solving a formation problem is to generate candidate solutions (or partial solutions) and test them against the given conditions or constraints. Examples of such tasks are *planning* and *design.* Many of the problems encountered by engineers fall between these two ends of the spectrum. A detailed discussion of the tasks mentioned above is beyond the scope of this work. The reader is referred to [29] for such a review.

An expert system typically consists of the following components, illustrated in Figure 1 [15]:

- Knowledge-Base. The knowledge-base consists of general facts and heuristic (rules of thumb) knowledge. A number of formalisms, such as production rules, frames (concepts) and semantic nets, are available for representing knowledge. The production rule representation has been extensively used in current KBES. In this approach, knowledge is represented as "IF-THEN" rules or "premise-action" pairs; the "action" is taken if the "premise" evaluates to be true. The knowledge-base may also be partitioned into knowledge levels in order to help organize the problem solving activities.

- Context. The context is a collection of symbols or facts that reflects the current state of the problem at hand. It consists of all the information generated during a particular program execution.

- Inference machine. The inference machine (Inference engine and inference mechanism are other terms commonly used instead of inference machine) monitors the execution of the program by using the knowledge-base to modify the context. Important problem solving strategies used in some existing systems are: means ends analysis, problem reduction, backtracking, least-commitment principle, control using agendas, and hierarchical planning. A detailed description of these strategies can be found in [22, 28].

The components discussed above form the kernel of most existing expert systems. In addition, there are three modules which are desired in any expert system: the user-interface, knowledge-acquisition and explanation modules. A more complete description of knowledge-based expert systems for civil

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专家在某一特定领域通过长期实践积累的知识。由于这些系统使用领域-specific知识，因此被称为基于知识的专家系统。Gaschnig 等人[12]将基于知识的专家系统定义为：“基于知识的专家系统是交互式计算机程序，整合了判断经验、经验法则、直觉及其他专业知识，以就各类任务提供专业化的建议。”

在KBES开发过程中所遇到的问题范畴，由推导任务与形成任务所界定[2]。在推导问题中，问题的条件被作为解决方案的一部分给出。这类问题通常通过利用已有知识和数据来补全求解路径加以解决。该范畴内常见的一些问题包括：解释、诊断、监控和预测。而形成问题则是由整个解决方案必须满足的条件或约束所提出的。解决形成问题的一种常用方法是生成候选方案（或部分方案），并将其与给定的条件或约束进行检验。此类任务的典型例子包括规划与设计。工程师所面临的许多问题介于上述两个极端之间。关于上述任务的详细讨论已超出本文的范围，读者可参阅文献[29]以获取相关综述。

专家系统通常由以下组件构成，如图1所示[15]：

- 知识库。知识库由一般性事实和启发式（经验法则）知识构成。目前已有多种形式化表示方法可用于知识的表达，如产生式规则、框架（概念）以及语义网络等。在当前的知识库专家系统中，产生式规则表示法得到了广泛应用。在这种方法中，知识被表示为“如果—那么”规则或“前提—动作”对；当“前提”条件为真时，即执行相应的“动作”。此外，为了便于组织问题求解活动，知识库还可按知识层次进行划分。

- 上下文。上下文是一组符号或事实的集合，用于反映当前所处理问题的状态。它包含了在特定程序执行过程中产生的所有信息。

- 推理机。推理机（在某些文献中也称作推理引擎或推理机制）通过利用知识库来修改上下文，从而监控程序的执行过程。一些现有系统中采用的重要问题求解策略包括：手段—目的分析、问题归约、回溯、最小承诺原则、基于议程的控制以及层次化规划。这些策略的详细说明可见文献[22, 28]。

上述所讨论的组件构成了大多数现有专家系统的内核。此外，任何专家系统还应具备三个模块：用户界面模块、知识获取模块和解释模块。关于土木工程领域的基于知识的专家系统的更为详尽的描述如下：

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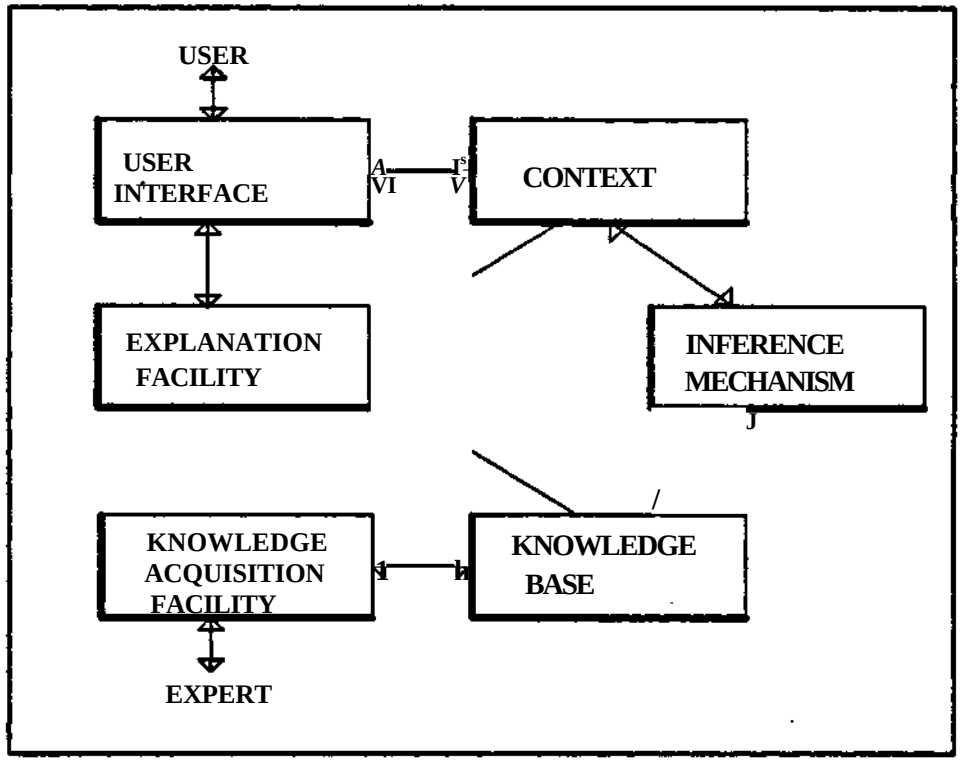



**Figure 1**: Schematic View of a KBES

engineering is **treated in [26].**

## OVERVIEW OF STRUCTURAL DESIGN PROCESS

The structural design process.starts with the definition of a need to transmit loads in space to a support or foundation, subject to constraints on cost, geometry and other criteria. The final product of the design process is the detailed specification of a structural configuration capable of transmitting these loads with the appropriate levels of safety and serviceability. The design process may be viewed as a sequence of three stages:

1. Preliminary design (conceptual design) involves the synthesis of a potential structural configuration satisfying a few key constraints. This stage of design may be further reduced to a series of subproblems. The first subproblem consists of defining the domain of feasible structural configurations applicable to the particular design at hand and choosing one or at most a few of these configurations. The second subproblem involves the formulation of specific constraints applicable to the configuration(s) chosen. This step may require an approximate analysis to estimate the response of the structure. The third subproblem requires a preliminary choice of parameters that define the

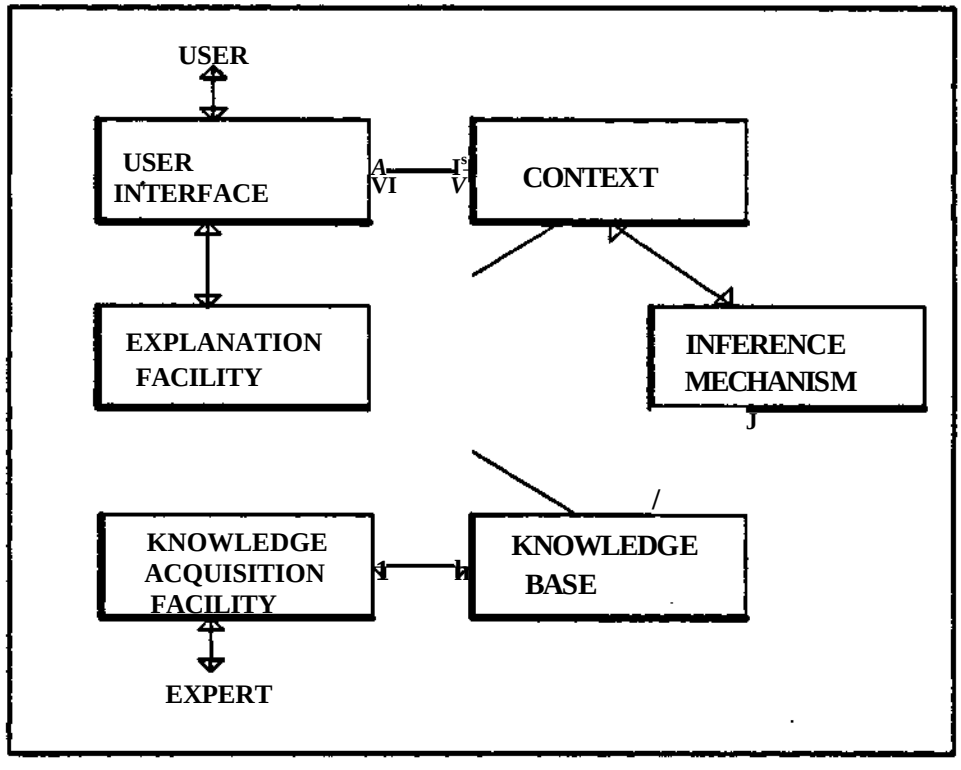



图1：KBES的示意图

工程学在文献[26]中得到了讨论。

结构设计过程概述

结构设计过程始于对在空间中将荷载传递至支承或基础这一需求的界定，同时需满足成本、几何形态及其他相关约束条件。设计过程的最终成果是对一种能够以适当的可靠性和使用性能传递这些荷载的结构构型所作的详细规定。该设计过程可被视为由三个阶段组成的序列：

1. 初步设计（概念设计）涉及对满足若干关键约束的潜在结构构型进行综合。该设计阶段可进一步分解为一系列子问题。第一个子问题在于确定适用于特定设计的可行结构构型集合，并从中选择一个或至多几个构型。第二个子问题涉及针对所选构型制定具体的约束条件，这一步骤可能需要通过近似分析来估算结构的响应。第三个子问题则要求对用于定义结构的参数进行初步选定。

configuration(s).

2. Analysis is the process of modeling the selected structural configuration and determining its response to external effects. Analysis may be viewed as the satisfaction of the functional constraints of equilibrium, compatibility and material stress-strain relationships. There are three subproblems in this stage. The structural configuration must first be transformed from a real structure to a mathematical model. This mathematical model is then analyzed. The final part is the interpretation of the results of the analysis in terms of the actual physical structure.

3. Detailed design is the selection and proportioning of the structural components such that all applicable constraints are satisfied. This stage again subdivides into a series of essentially hierarchical subproblems, such as detailing the main structural components (beams, columns, etc.) followed by detailing of their subcomponents (connections, reinforcement, etc.). Within each subproblem, there is a further subdivision into selection based on some controlling constraints (e.g., toad-earring capacity or buckling) followed by the evaluation of secondary constraints (e.g., local buckling or crippling).

There may be significant deviations between the properties of components assumed at the analysis stage and those determined at the detailed design stage, which would necessitate a reanalysis. Other major and minor cycles of redesign may also occur. The process continues until a satisfactory (or optimal) design is obtained. The *conceptualize-ana/yze-detail* cycle is typical of many design paradigms.

The following sections present example applications of expert systems to each of the three stages of structural design.

## PRELIMINARY DESIGN

This stage of design is illustrated by a KBES that aids in the preliminary structural design of high rise buildings (HI-RISE) [18]. HI-RISE is implemented in PSRL [24], a frame based production system language being developed at Carnegie-Mellon University. PSRL is a general purpose programming language that provides the constructs available in a production system language and a schema representation language.

The knowledge-base of HI-RISE contains declarative and procedural knowledge. The declarative knowledge is stored in lists and represents a physical hierarchy of known structural types. For example, one physical hierarchy, used for configuring a lateral load resisting system, is given below.

- 3D-subsystems: core, tube

配置。

2. 分析是指对所选定的结构构型进行建模，并确定其对外部作用的响应过程。分析可被视为对平衡、协调以及材料应力-应变关系等功能约束的满足。该阶段包含三个子问题：首先，需将实际结构转化为数学模型；随后对这一数学模型进行分析；最后，将分析结果与实际物理结构相对应并加以解释。

3. 详细设计是指对结构构件进行选型与配比，以确保满足所有适用的约束条件。该阶段进一步细分为一系列本质上呈层级关系的子问题，例如先对主要结构构件（梁、柱等）进行详图设计，随后再对其子构件（连接节点、钢筋配置等）进行详图设计。在每个子问题中，又可进一步划分为基于某些控制性约束条件的选型（如承载力或屈曲），随后是对次要约束条件的校核（如局部屈曲或压溃）。

在分析阶段所假设的构件性能与详细设计阶段确定的构件性能之间可能存在显著偏差，这将导致需要重新进行分析。此外，还可能出现其他主要或次要的再设计循环。这一过程将持续进行，直至获得令人满意的（或最优的）设计方案。概念化—分析—细化的循环是许多设计范式中的典型模式。

以下各节将分别介绍专家系统在结构设计三个阶段中的应用实例。

## 初步设计

该设计阶段通过一个基于知识的工程系统（KBES）加以说明，该系统有助于高层建筑（HI-RISE）的初步结构设计[18]。HI-RISE是在PSRL[24]中实现的，PSRL是一种基于框架的产生式系统语言，正在卡内基梅隆大学开发。PSRL是一种通用编程语言，同时具备产生式系统语言和模式表示语言所支持的构造机制。

HI-RISE的知识库包含陈述性知识和程序性知识。陈述性知识以列表形式存储，表征已知结构类型的物理层次结构。例如，用于配置抗侧力体系的一个物理层次结构如下所示。

- 3D子系统：核心、管状结构

- 2D-subsystems: solid shear wall, braced frame, rigidly connected frame
- material: steel, reinforced concrete, prestressed concrete

The procedural knowledge is represented by production rules and LISP functions. The procedural knowledge is organized into sections or *knowledge modules*. The top level division of the knowledge modules concerns the design of two functional systems: design of the lateral load resisting system, and design of the gravity load resisting system. The design of each functional system is further decomposed into the following knowledge modules.

- Synthesis. A set of alternative configurations are synthesized for the functional system under consideration. The synthesis is performed as a depth first search through the appropriate physical hierarchy in the knowledge-base. The search space is pruned by constraint elimination using heuristic elimination rules.
- Analysis. The major purpose of the analysis module is to evaluate system feasibility. This is done by the formulation and evaluation of one or more feasibility constraints. Analysis provides one ingredient of the constraint, namely the required load capacity of the system components. Parameter selection provides the remaining ingredients required for constraint evaluation.
- Parameter selection. The parameters of the system are initially selected using heuristics. These initial parameters are used to evaluate the feasibility constraints. If a constraint is violated, some heuristic recovery rules are applied to revise the parameters. Once satisfactory parameters are selected, i.e., all applicable constraints are satisfied, the alternative is evaluated.
- Evaluation. Alternative configurations are evaluated with a linear evaluation function. There is a distinct evaluation function for each of the functional systems. The variables in the function are features of the system that may be quantified.
- System selection. All structurally feasible configurations are presented to the user indicating which configuration has been determined to be the "best", selected as the configuration with the minimum value assigned by the evaluation function. The user may either accept the recommended design or override the decision of HI-RISE and choose one of the other structurally feasible systems.

The context tree of HI-RISE represents the solution to the design problem. The tree is decomposed into a functional plane and a physical plane. The functional plane contains two subtrees: a representation of the lateral load resisting system and a representation of the gravity load resisting system. The physical plane contains a hierarchy of the structural subsystems that define the functional systems. The physical plane of the context tree is built up from the physical hierarchy contained in the knowledge-base.

The problem solving strategy in HI-RISE is developed using the inference mechanism provided by

- 2D subsystems: solid shear wall, braced frame, rigidly connected frame
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PSRL. PSRL provides several conflict resolution strategies; one strategy is selected for each rule set. The following strategies are used by HI-RISE:

- Fire-cycle. The first rule whose left hand side (LHS) is satisfied is fired, i.e, its right hand side (RHS) is evaluated. The next cycle starts at the beginning of the rule set and fires the first rule whose LHS is satisfied. Cycles continue until there are no rules in the rule-set whose LHS is satisfied.

- Fire-sequence. As in fire-cycle, the first rule with a satisfied LHS is fired. Subsequent cycles begin with the rule immediately following the previously fired rule. Cycles continue until the LHS of the last rule in the rule set is matched with working memory and it is not satisfied.

- Fire-one. Again the first rule that is satisfied is fired. There is only one cycle in this variation of conflict resolution.

Rule sets are used to develop a hierarchical planning strategy, implemented by decomposing the design problem into subprobiems. The subprobiems are solved in a fixed order; first the lateral load resisting system is designed followed by the design of the gravity load resisting system. The design of the functional systems are decomposed into subprobiems corresponding to the knowledge modules and are solved in a fixed **order.**

## ANALYSIS

Analysis is the process of determining the response of a fully specified structure to its environment. The response quantities of interest are the stresses and stress resultants (forces and moments) in various regions or components and the deflections of various points. The environment describes the loads acting on the structure, including equivalent loads produced by thermal effects, material changes such as shrinkage and creep, or differential displacements. A fully specified structure is one whose topology, material properties and boundary conditions have all been previously described.

There are literally hundreds of general-purpose structural analysis programs, most of them based on the finite element method, available to the structural designer. Because of the comprehensiveness of the programs, it may take a user a year or more to learn to effectively and efficiently use the common options and capabilities of a large structural analysis program. The total set of capabilities of a large structural analysis program is not likely to be exercised by any single user during his entire professional career.

Knowledge-based expert systems can provide substantial assistance in translating the engineer's knowledge of *what* he wishes to learn from the analysis to *how* to specify this information to an

PSRL。PSRL 提供多种冲突解决策略，每组规则均会选用其中一种策略。

HI-RISE采用以下策略：

- 火循环。首先触发第一个其左部（LHS）满足的规则，即对其右部（RHS）进行求值。随后，下一个循环从规则集的起始位置开始，触发第一个其左部满足的规则。循环持续进行，直至规则集中不再存在左部满足的规则。

- 火序列。与火循环类似，首先触发左部条件（LHS）满足的第一条规则。随后的各轮次均从紧接在上一轮次所触发规则之后的规则开始执行。循环持续进行，直至规则集中最后一条规则的左部条件与工作内存匹配但不成立为止。

- 触发规则一。再次，第一个被满足的规则被触发。在该冲突消解策略的变体中，仅包含一个循环。

规则集用于构建一种分层规划策略，该策略通过将设计问题分解为子问题来实现。子问题按照既定顺序求解：首先设计抗侧力体系，随后设计抗重力体系。功能系统的设计则被分解为与知识模块相对应的子问题，并按固定顺序予以求解。

## 分析

分析是指确定一个已完全定义的结构对其所处环境的响应的过程。关注的响应量包括各区域或构件中的应力及应力分量（内力和弯矩），以及各点的位移。环境则描述作用于结构上的荷载，其中包括由温度效应、材料特性变化（如收缩与徐变）或差异位移所产生的等效荷载。所谓“已完全定义的结构”，是指其拓扑构型、材料性能及边界条件均已事先给定的结构。

结构设计师可使用的通用结构分析程序多达数百种，其中大多数基于有限元法。由于这些程序功能全面，用户可能需要一年甚至更长时间才能熟练、高效地掌握大型结构分析程序的常用选项与功能。即便在其整个职业生涯中，任何一位用户也很难将大型结构分析程序的全部功能都充分运用到实践中。

基于知识的专家系统能够为将工程师希望从分析中获取的知识转化为如何向分析工具提供这些信息的具体说明提供重要支持。

analysis program. Such assistance can substantially elevate the role of an engineer from that of an analyst to that of a modeller, concerned with the choice of the appropriate analytical model and with the interpretation of the analysis results. Furthermore, an expert system knowledge-base can serve as a pool of the accumulated knowledge and modeling expertise of individuals within design organization or even larger professional groupings.

## SACON - Structural Analysis CONsultant

An early expert system addressing some aspects of an *automated consultant* to advise non-expert engineers in the use of a general structural analysis program was SACON [4]. SACON was implemented in the EMYCIN system, which is essentially the MYCIN program with the medical diagnosis knowledge removed. This section provides a brief summary of [4].

The structural mechanics knowledge-base of SACON consists of:

- rules for inferring analysis strategies, consisting of the identification of the most appropriate analysis class to be performed and associated analysis recommendations;
- rules for inferring controlling stress, deflection and nonlinear behavior of substructures; and
- mathematical models, in the form of *procedural attachments* for estimating non-dimensional stress and deflection bounds for each substructure, based on its boundary conditions and loading.

The rule base of SACON is encoded as inference rules indicating an action to be performed if the conditions are fulfilled. Premises are conjunctions of clauses, each clause being of the form

```
<predicate function> <object> <attribute> <value>.
```

The effect of the actions is to add or modify a fact in the context, where each fact is represented as a 4-tuple in the form

```
<object> <attribute> <value> <certainity factor>.
```

An example rule for stress behavior is shown below:

```
IF      The material composing the sub-structure is one of:
        the metals, AND
        The analysis error, in percent, that is tolerable
        is between 5 and 30, AND
        The non-dimensional stress of the sub-structure is
```

分析程序。此类辅助工具能够显著提升工程师的角色定位，使其从单纯的分析者转变为建模者，重点关注恰当的分析模型选择以及分析结果的解释。此外，专家系统知识库还可作为设计机构乃至更大范围的专业群体中个人所积累的知识与建模经验的集合平台。

SACON - 结构分析咨询顾问

早期的一个专家系统——SACON[4]——旨在解决自动化顾问的部分功能，为非专业工程师在使用通用结构分析程序时提供指导。SACON是在EMYCIN系统上实现的，该系统本质上是去除了医学诊断知识的MYCIN程序。本节将对文献[4]作简要概述。

SACON的结构力学知识库由以下内容组成：

- 用于推断分析策略的规则，包括确定应执行的最适宜分析类别及相应的分析建议；
- 用于推断子结构的控制应力、变形及非线性行为的准则；以及
- 数学模型以程序化附录的形式呈现，用于根据各子结构的边界条件和荷载情况，估算其无量纲应力和变形的上限值。

SACON的规则库被编码为推理规则，用于在条件满足时指示应执行的操作。前提是由若干子句构成的合取式，每个子句的形式为

<谓词 函数> <对象> <属性> <值>。

这些操作的效果是在上下文中添加或修改一个事实，其中每个事实均以一个四元组的形式表示，其形式为

**<object> <attribute> <value> <certainity factor>.**

应力行为的一个示例规则如下所示：

如果　构成该子结构的材料为金属之一，并且可容忍的分析误差（以百分比计）介于5%至30%之间，同时该子结构的无量纲应力为

```
          greater than 0.9. ANO
          The number of cycles the loading is to be applied is
          between 1000 and 10000
THEN      It is definite (1.0) that fatigue is one of the stress
          behavior phenomena in the substructure
```

SACON dynamically constructs a context tree from a fixed hierarchy to represent the relationships between the objects. The variable bindings in a rule are determined by the context in which the rule is fired.

The inference mechanism uses backward chaining that produces a depth-first search of a goal tree. In order to reach a goal, i. e., establish a fact, SACON retrieves all rules that mention that fact in their action and invokes each one in turn, evaluating its premise to see if the conditions specified have been met If any fact needed to establish the premise is not in the context, a subgoa! is set up to establish it and the process recurs. If there are no relevant rules to establish a fact, the user is asked for the value. The generalized subgoai is "determine the value of <attribute>[H], rather than "determine whether <attribute> is equal to a specific <value>", thus tending to group related questions together.

Another KBES developed on the same lines as SACON is SesCon [10], a prototype system developed as a front end for the use of the Seasam-69 structural analysis package.

## EXTENSIONS AND POSSIBLE IMPLEMENTATION

An extension of a *passive* analyst consultant proposed in [4] would be to integrate the expert system with one or more analysis programs to produce a *dosed loop* system as illustrated in Figure 2. In such a system, the expert system's recommendations (subject to user intervention) would be translated into specific input data, the analysts program executed, and its results returned to the expert system for comparison with its initial predictions and recommendations. If the expert system discovered modeling inconsistencies, it could recommend alternate analysis strategies or model refinements.

An earlier attempt was made by Rivlin et al. [23] to develop such a system. The system, written in FORTRAN, has a finite element KB and has an interface with the analysis program. The system does not have a [M]user friendly" natural interface and is very difficult to use. However, recent developments in user-system interfaces [16] and expert system tools, such as EXPERT [31], [32], provide the necessary tools to build the closed loop sysem shown in Figure 2.

大于0.9。ANO 加载循环次数介于1000至10000之间。

然后　　可以明确（1.0）地指出，疲劳是下部结构中应力行为现象之一。

SACON 通过从预定义的层次结构动态构建上下文树，以表示对象之间的关系。规则中的变量绑定由该规则被触发时的上下文决定。

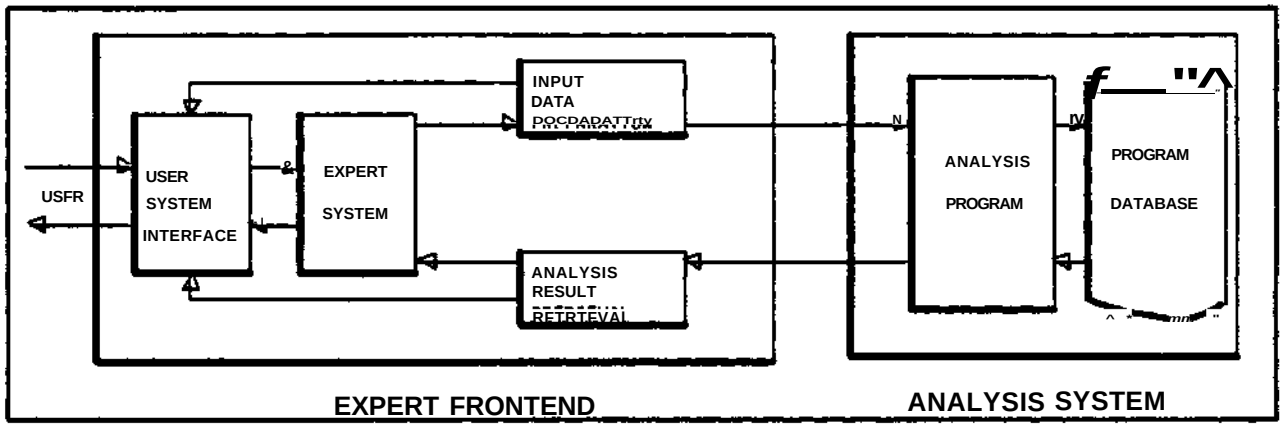
推理机制采用逆向链式推理，以深度优先方式搜索目标树。为达成某一目标，即确立某一事实，SACON 会检索所有在其动作部分提及该事实的规则，并依次调用每一条规则，评估其前提条件是否已满足。若确立前提所需的某项事实不在当前上下文中，则会设立一个子目标以进一步确立该事实，并递归执行上述过程。若不存在可用于确立该事实的相关规则，则系统将向用户询问其值。广义的子目标表述为“确定<属性>H的值”，而非“确定<属性>是否等于特定的<值>”，从而倾向于将相关问题归并在一起。

另一项与SACON采用相同思路开发的KBES是SesCon[10]，该系统是一个作为Seasam-69结构分析软件包前端使用的原型系统。

扩展与可能的实现

对文献[4]中提出的被动式分析顾问的一种扩展方案是，将专家系统与一个或多个分析程序集成，从而构建如图2所示的闭环系统。在这样的系统中，专家系统的建议（经用户干预后）将被转化为具体的输入数据，随后执行分析程序，并将其结果返回给专家系统，以与其最初的预测和建议进行比对。若专家系统发现模型存在不一致之处，可推荐备选的分析策略或对模型进行优化调整。

Rivlin 等人[23]曾尝试开发此类系统。该系统采用 FORTRAN 语言编写，内置有限元知识库，并与分析程序实现了接口连接。然而，该系统缺乏“用户友好”的自然交互界面，使用起来十分不便。不过，近年来在人机交互界面[16]以及专家系统工具（如 EXPERT[31, 32]）方面的进展，为构建图 2 所示的闭环系统提供了必要的技术支撑。




**Figu re 2: A Conceptual View of an Integrated Analysis System**

A complete structural analysis assistant may eventually contain several levels of modeling and model interpretation knowledge. A hierarchy of knowledge sources may be organized as follows:

- modeling strategies: significant modes of behavior (static vs. dynamic, linear vs. nonlinear, etc.) to consider; likely failure modes or limit states (yielding, buckling, etc.) applicable to the structure;
- global idealization: appropriate "granularity" or level of modeling; appropriate subdivisions into substructures;
- local idealization: applicable or appropriate mesh size, element types, loading and boundary condition representations;
- result interpretation: appropriateness of models and idealizations chosen; meaningful aggregations of data to discern patterns and trends; ways of evaluating sensitivity of the model;
- evolutionary strategies: appropriate initial models and idealizations; ways of interpreting results and progressively refining the model;
- efficiency issues: types of models that are efficiently handled by a given program; ways of overcoming program inefficiencies; and
- program details: rules for translating the model into the input language of a particular program.

The hierarchical structuring would result in a more efficient use of the knowledge-base.

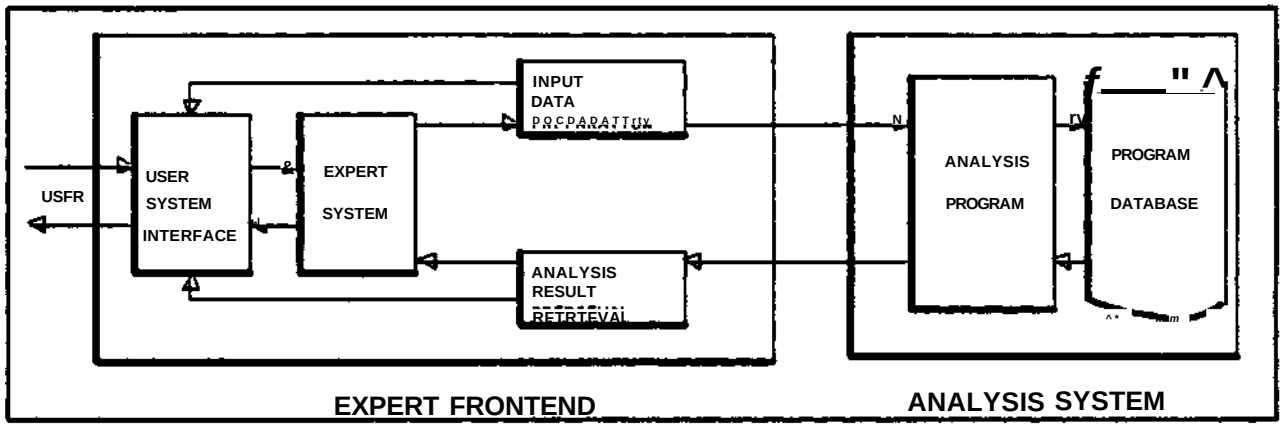



图2：集成分析系统的概念视图

一个完整的结构分析辅助系统最终可能包含多个层次的建模与模型解释知识。知识源的层级结构可组织如下：

•建模策略：需考虑的重要行为模式（静态与动态、线性与非线性等）；适用于该结构的可能失效模式或极限状态（屈服、屈曲等）。

•全局理想化：合适的“粒度”或建模层次；适当的子结构划分；

•局部理想化：适用或恰当的网格尺寸、单元类型、载荷及边界条件的表达方式；

•结果解释：所选用模型与理想化的合理性；用于识别模式与趋势的数据有意义的聚合方式；评估模型敏感性的方法；

•进化策略：恰当的初始模型与理想化假设；结果的解释方式及模型的逐步精炼；

•效率问题：给定程序能够高效处理的模型类型；克服程序效率低下的方法；以及

•程序细节：将模型转换为特定程序输入语言的规则。

分层结构化将有助于更高效地利用知识库。

## DETAILED DESIGN

The detailed design of structural components is governed by many constraints, some mandated by design specifications and building codes, others representing limitations and "design styles" of individual design firms. For example, the two production rules

```
IF      Actual stress f <= allowable stress F
THEN    Stress constraint satisfied
```

and

```
IF      Section is compact, AND
        Not hybrid girder or made of A514 steel
THEN    Allowable stress in bending Fb * 0.66 Fy
```

represent code constraints based in sections 1.5 and 1.5.1.4.1, respectively, of the AISC Steel Design Specification [1]. On the other hand, the decision on whether to use a compact section or not may be a heuristic based in the firm's "design style" of handling specific situations.

The incorporation of specification constraints mandated by codes into computer aided design programs is a two-step process [8]:

1. representing the constraints in a form that can be manipulated by computer programs; and

2. converting passive conformance checking constraints into an active design sequence.

The first problem was addressed by Fenves and others [14,20], by representing provisions in the form of networks of decision tables; the processing of these tables was implemented in a computer program by Stirk [30]. The second issue was addressed in [17]. An alternative or extension of the above is to represent these constraints in the knowledge-base of a KBES (Research in this area is also actively being pursued in U. K. and Australia, see [13] for further details).

### SPECON -SEEcification CONsultant

Specifications formulated as decision tables can be recast into expert system representational schemes, such as frames and production rules. The application of expert systems to provision specification is illustrated by using the production rule approach.

SPECON is a small but representative knowledge-based system intended to aid the engineer in checking of structural steel elements for conformance with the AISC Steel Design Specification [1]. It

详细设计

结构构件的详细设计受到诸多约束的制约，其中一些是由设计规范和建筑法规所强制规定的，另一些则体现了各设计单位的限制条件及“设计风格”。例如，两条生产规则

如果实际应力 f ⇐ 小于或等于许用应力 F，则应力约束满足。

a且若截面为紧凑截面，并且非组合梁或由A514钢制成，则允许的弯曲应力$F_b$ * 0.66 $F_y$。

分别代表依据AISC《钢结构设计规范》[1]第1.5节和第1.5.1.4.1节所规定的截面约束条件。另一方面，是否采用紧固型截面的决策可能是一种基于企业处理特定情形时的“设计风格”的启发式判断。

将规范所规定的约束条件纳入计算机辅助设计程序是一个两步过程[8]:

1. 以一种可被计算机程序处理的形式表示约束条件；以及

2. 将被动式一致性检查约束转化为主动式设计序列。

第一个问题已由Fenves等人[14,20]通过将各项规定以决策表网络的形式加以表示来解决；这些决策表的处理过程则由Stirk在计算机程序中予以实现[30]。第二个问题已在文献[17]中得到解决。上述方法的一种替代方案或扩展是将这些约束条件表示为基于知识的专家系统（KBES）的知识库的一部分（该领域的研究在英国和澳大利亚也正在积极开展，详见文献[13]）。

SPECON - 规范顾问

以决策表形式表述的规范可以重新表述为专家系统的表示框架，例如框架和产生式规则。通过采用产生式规则方法，说明了专家系统在规范制定中的应用。

SPECON 是一个规模虽小但具有代表性的基于知识的系统，旨在帮助工程师对钢结构构件是否符合 AISC 钢结构设计规范[1]进行校核。它

was implemented in LISP [33], a list processing language, and OPS5[11], a production system language.

The organization of the system is similar to that of MYCIN [25] and SACON; it consists of three main modules: the knowledge-base, the context, and the inference machine. The essential difference between SPECON and other expert systems is the flexibility provided to the user to alter numerical values of design parameters until the hypothesis is satisfied, i.e., a component satisfying the constraints is selected.

The knowledge-base is divided into two levels. The first level consists of production rules which identify the applicable constraints. The second level is made up of the specific design constraints, in the same form as discussed in the section on preliminary design. The second level constraints are called from first level rules whenever applicable preconditions are satisfied.

Illustrative rules representing Section 1.5.1.4 of [1] are shown below:

```
First level * set up constraints:

RULE 001
   IF    Section contains unstiffened elements, AND
         Constraint for unstiffened elements satisfied
   THEN  Unstiffened section requirement satisfied


RULE 013
   IF    Constraint for doubly symmetric section satisfied, AND
         Unstiffened section requirement satisfied
   THEN  Allowable bending stress (Fb) is 0.66 times
         the  yield stress (Fy)

RULE 029
   IF     Actual bending stress (fb) < allowable bending stress (Fb)
   THEN   Bending stress requirement  satisfied

Second level-constraint evaluation:

RULE 032
   IF     bf/(2*tf) < 65/Sqrt(Fy)
   THEN  Constraint for unstiffened element satisfied

 {bf and tf are the width and thickness of the flange}
```

The context in the LISP implementation consists of four list structures that keep track of the various facts generated in a particular consultation. In the OPS5 implementation, this is reflected in the working memory.

该方法分别在LISP（一种列表处理语言）和OPS5（一种产生式系统语言）中得以实现。

该系统的组织结构与MYCIN[25]和SACON相似，主要由三个模块组成：知识库、上下文和推理机。SPECON与其他专家系统之间的本质区别在于，它为用户提供了修改数值参数的灵活性。

l

调整设计参数的取值，直至假设条件得到满足，即选定一个满足约束条件的部件。

知识库被划分为两个层次。第一层由用于识别适用约束的产生式规则组成。第二层则由具体的设计约束构成，其形式与初步设计章节中所讨论的一致。当相应的前提条件得到满足时，第二层的约束将由第一层的规则调用。

图例 表示[1]中第1.5.1.4节的激活规则如图所示。 如下：

第一层 * 设置约束条件：

规则001：如果截面包含非加劲板件，且非加劲板件的约束条件得到满足
，则非加劲截面的要求得到满足。

规则013：若双重对称截面的约束条件满足，且无加劲截面的要求亦得到满足，则允许弯曲应力($F_b$)为屈服应力($F_y$)的0.66倍。

规则029：若实际弯曲应力（$f_b$）<许用弯曲应力（$F_b$），则弯曲应力要求满足。

第二级约束评估：

规则032：若 $b_f/(2*t_f) < 65/\sqrt{(F)}$ 成立，则非加劲构件的约束条件满足
。

{$b_f$ 和 $t_f$ 分别为翼缘的宽度和厚度}

在LISP实现中，上下文由四个列表结构组成，用于记录在特定咨询过程中生成的各种事实。在OPS5实现中，这一点则体现在工作存储中。

The control strategy implemented in the inference machine is similar to MYCIN's backward chaining strategy. Essentially, it involves starting from the top hypothesis, *Bending stress requirement satisfied,* and checking the facts in the context to see if it can be supported from these facts. If the facts are not available, then the preconditions that are needed for the hypothesis are set up as sub-goals. For example, the condition part of Rule 029 is set up as a sub-goal if it is not found in the context

The explanation module of the system can answer questions, such as

- how a certain hypothesis is deduced, and
- why a particular question was asked.

The user can ask the system any time during the consultation to explain its reason for asking a certain question by typing "why". If the user types "how-did-you-deduce" after a consultation session is over, then the system would explain its reasoning process for any desired **fact.**

## EXTENSIONS AND POSSIBLE IMPLEMENTATIONS

A complete detailed design consultant may incorporate the following additional features:

- larger knowledge-base, incorporating all the mandated constraints of codes and specifications such as [1], possibly including the reason for each constraint;
- soft constraints, such as those representing office practices and "design styles" of individual organizations;
- external constraints, such as those representing good field practices, and relative costs of details;
- meta-rules, such as heuristics for choosing which constraints to apply at which stage of design; and
- design rules f17] to provide bounds on variables such that controlling constraints are satisfied.

The practical implementation of such an expert system must be closely integrated with a database management system which maps the context onto a project file containing the description of the design. For example, the bending stress constraint illustrated in the previous section must eventually be applied to every element in the structure; in order to evaluate the condition "section contains unstiffened elements", the database must be accessed to obtain the composition of each particular section, etc. A mechanism for treating such constraints is presented in [9]. Currently, this topic is an active area of research at Carnegie-Mellon University.

推理机中所采用的控制策略类似于MYCIN的逆向推理策略。其核心在于从顶层假设“弯曲应力满足要求”出发，通过检查上下文中的事实来判断该假设是否能够得到这些事实的支持。若相关事实缺失，则将该假设所需的先决条件作为子目标加以设定。例如，若规则029的条件部分在上下文中未被发现，则将其作为子目标进行处理。

该系统的解释模块能够回答诸如……之类的问题。

- 某个假设是如何被推导出来的，以及
- 为什么提出了某个特定问题。

用户可在咨询过程中随时通过输入“why”来请求系统解释其提出某一问题的原因。若在咨询会话结束后，用户输入“how-did-you-deduce”，则系统将针对任意指定的事实说明其推理过程。

扩展与可能的实现

一份完整的详细设计咨询方案可能包含以下附加功能：

- 更大的知识库，其中整合了诸如[1]之类的规范与标准所规定的全部约束条件，并可能包含每条约束的依据；
- 软约束，例如那些代表各组织的办公惯例和“设计风格”的约束；
- 外部约束条件，例如那些代表良好现场作业规范的约束，以及各细节项的相对成本；
- 元规则，例如用于确定在设计的哪个阶段应用哪些约束的启发式方法；以及
- 设计规则[17]用于为变量设定约束范围，以确保关键约束条件得到满足。

此类专家系统的实际实现必须与数据库管理系统紧密集成，该系统将上下文映射到包含设计描述的项目文件中。例如，前一节中所阐述的弯曲应力约束最终必须施加于结构中的每一个构件；为了评估“截面包含非加劲构件”这一条件，必须访问数据库以获取每个特定截面的组成信息，等等。关于此类约束的处理机制已在文献[9]中予以阐述。目前，这一课题仍是卡内基梅隆大学的一个活跃研究方向。

## INTEGRATED SYSTEM

In the previous sections, applications of expert systems in the three stages of structural design were presented. In actual design of complex structures these three stages are performed by different experts. For example, architects, space planners, and structural engineers are normally jointly responsible for the preliminary design, while the detailed design is accomplished by teams of structural engineers, detailers, and draftsmen. Furthermore, each of these tasks are divided into a number of subtasks; dividing the design task itself demands considerable experience. Hence the design task requires a good deal of coordination among different experts. Based on these considerations, a *conceptual* view of an integrated design system that utilizes the concept of cooperation of different experts is proposed. The system is intended for a single user, although it provides an excellent paradigm for the development of a multiuser system. The input to the system consists of information provided by space planning, including the number of stories, the plan of the building, and other pertinent spatial constraints.

The overall organization of the system is similar to that of HEARSAY-II [7] and HASP/SIAP [19] systems: it consists of a number of *knowledge-modules* that communicate through a *blackboard*, as shown in Figure 3, and controlled by an *inference mechanism*. These components are discussed below.

The knowledge-base consists of a number of knowledge modules (KMs). Some KMs incorporate both *textbook* and *heuristic* (surface) knowledge, while other KMs comprise *fairly deep* knowledge (Surface knowledge consists mainly of *production rules* encoding empirical associations based on experience, while fairly deep knowledge is comprised of algorithmic procedures). These KMs form a hierarchy of four levels: the *strategy, activation, specialist,* and *resource levels*. The knowledge levels are briefly described below.

- Strategy level KMs analyze the current solution state to determine the course of next action. This level supplies the knowledge required for *focusing attention* on a certain part of the solution. Knowledge from this KM is used by the blackboard monitor to fire the appropriate activator.

- Activation level KMs know when to use the Specialist KMs. The task of KMs at this level is to invoke the appropriate specialists for a given strategy. The system may incorporate both activators available in the HASP system - *the event driver* and *the goal driver*. The event driver invokes appropriate Specialists based on a certain event or group of events. The goal driver acts on those KMs which are responsible for putting a potential design

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alternative in the blackboard. For example, the event driver would invoke the KM for analyzing the structure if all the required input data is available, while the goal driver would set up the task of analysis as a goal and would try to obtain the input data from the user or other KMs. Essentially, the event and goal drivers are used to control the sequence of execution of the knowledge modules (KMs). This type of knowledge is called *process control knowledge* (see the section on the inference mechanism for a different type of control structure).

- Specialist KMs are used to assess the current state of the design. Some KMs are comprised of pure production rules, while other KMs invoke analytical models of knowledge from the Resource plane. The KMs at this level are comprised of smaller independent modules which are called knowledge sources (KSs). There are two types of KSs: KSs unique to a certain KM and KSs common to several KMs. These KSs are activated by appropriate KMs. Each KM at the Specialist level will be specified in four parts [6]:

  1. conditions under which it is to be activated;

  2. kinds of changes it makes to the blackboard;

  a conditions under which KMs at the Resource level are activated; and

  4. conditions under which KSs are activated.

  Further details on the individual KMs, shown in Figure 3, are presented in [27].

- Resource level KMs contains the knowledge required for analysis and design. Typical KMs at this level are: analyzers, geometric modellers, codes, catalogs of available components, data base management systems, costing procedures, etc.

The blackboard or context consists of two planes: the functional plane and the physical plane. The functional plane is subdivided into two functional subsystems: lateral load resisting system, and gravity load resisting system. The hierarchy of levels in the physical plane was discussed in the section on preliminary design.

The main units in the blackboard are hypotheses or guesses. For example, a possible 3-0 hypotheses such as a *Core* or *Tube-in-tube* may exist in the blackboard. The hypotheses are primary guesses about particular aspects of the problem and may be related through structural links. For example, the gravity load functional level can have many alternative configurations which are linked through an is-aJt (is an alternative) link. The hypothesis structure formed from the most up-to-date information is known as the *current best design* (CBO). Constraints play an important role in forming the CBO. The interaction between various levels through constraints is addressed elsewhere [18].

The inference mechanism consists of two main components: *the agenda,* and *the monitor.* The agenda keeps track of all the events in the blackboard and calculate the priority of execution of tasks

在黑板中存在一种替代方案。例如，当所有必需的输入数据均已就绪时，事件驱动器将调用用于结构分析的知识模块；而目标驱动器则会将该分析任务设定为目标，并尝试从用户或其他知识模块处获取所需的输入数据。本质上，事件驱动器和目标驱动器用于控制知识模块的执行顺序。此类知识被称为过程控制知识（关于另一种类型的控制结构，请参见推理机制一节）。

- 专家级知识模块用于评估设计的当前状态。部分知识模块由纯产生式规则构成，而另一些知识模块则调用资源层中的分析型知识模型。该层级的知识模块由若干较小的独立模块组成，这些模块被称为知识源（KS）。知识源分为两类：一类为特定知识模块所独有的知识源，另一类为多个知识模块共用的知识源。这些知识源由相应的知识模块激活。专家级的每个知识模块将按四个部分进行规范描述[6]。

1. 其被激活的条件；

2. 其对黑板所做的两种改变；

a 资源级别知识模块被激活的条件；以及

4. KSs被激活的条件。

图3中所示的各个知识模块的进一步详细信息见文献[27]。

- 资源级知识模型包含分析与设计所需的知识。该层级的典型知识模型包括：分析工具、几何建模器、代码库、可用元件目录、数据库管理系统、成本核算流程等。

黑板图或结构体系由两个层面构成：功能层面和物理层面。功能层面进一步细分为两个子系统：抗侧力体系和抗重力体系。物理层面的层次划分已在初步设计部分予以阐述。

黑板中的主要单元是假设或猜想。例如，黑板中可能存在一种可能的3-0型假设，如核心筒或管中管结构。这些假设是对问题特定方面的初步猜测，并可通过结构性链接相互关联。例如，重力荷载功能层可以有多种备选构型，这些构型通过“is-a”（即“是一种”）链接相互关联。由最新信息形成的假设结构被称为当前最优设计方案（CBO）。约束在形成CBO的过程中起着重要作用。各层级之间通过约束的交互作用已在其他文献中予以阐述[18]。

推理机制由两个主要组成部分构成：议程和监控器。议程负责跟踪黑板中的所有事件，并计算任务的执行优先级。

that are generated as a result of activation of the KMs. The activation and interaction between various levels is accomplished by the blackboard monitor. In the proposed system, the agenda is implemented as a queue and the priority of execution may be set by the user; the user will have considerable control over the agenda. The monitor performs the following tasks:

- check the blackboard for appropriate KMs to fire; and
- take the activity with the highest priority from the agenda and execute it.

The activation of a certain KM depends on the result of actions of other KMs. For example, the KM for detailed design stage would be activated after the analysis is complete. This can be cast into the following production rule:

```
IF    The analysis of structure is complete, AND
      Detailed design is  required
THEN  Activate the knowledge-module for detailed design
```

Similar production rules can be used to provide an alternate solution to the problem of interaction of multidisciplinary design databases, described by Eastman [5].

Both the data-driven and goal-driven strategies are to be incorporated in the system. For example, if a KM generates a surface element, such as a grid, from linear elements (beams), then the hypothesis about the surface element is data-driven. On the other hand, if the type of surface element, such as the grid, is hypothesized in the light of domain knowledge or suggested by use, then the hypothesis formation is goal driven (Note that in the goal-driven case the hypothesis, i.e., grid, is checked for correctness and consistency from knowledge about linear beam elements which further depend on the material). The data-driven and goal-driven strategies act on the hypothesis structure and are termed *hypothesis-control* strategies.

## CONCLUSIONS

A number of applications of expert systems to structural design have been described. A conceptual view of an integrated structural design system using a knowledge-based approach was presented. The objective of these systems is to complement the extensive algorithmic use of computers in structural engineering by augmenting the engineer's problem solving capabilities in areas characterized by heuristic expertise. Prototype implementations of all the applications described above can be developed using existing expert system frameworks such as OPS5, PSRL, KAS [21],

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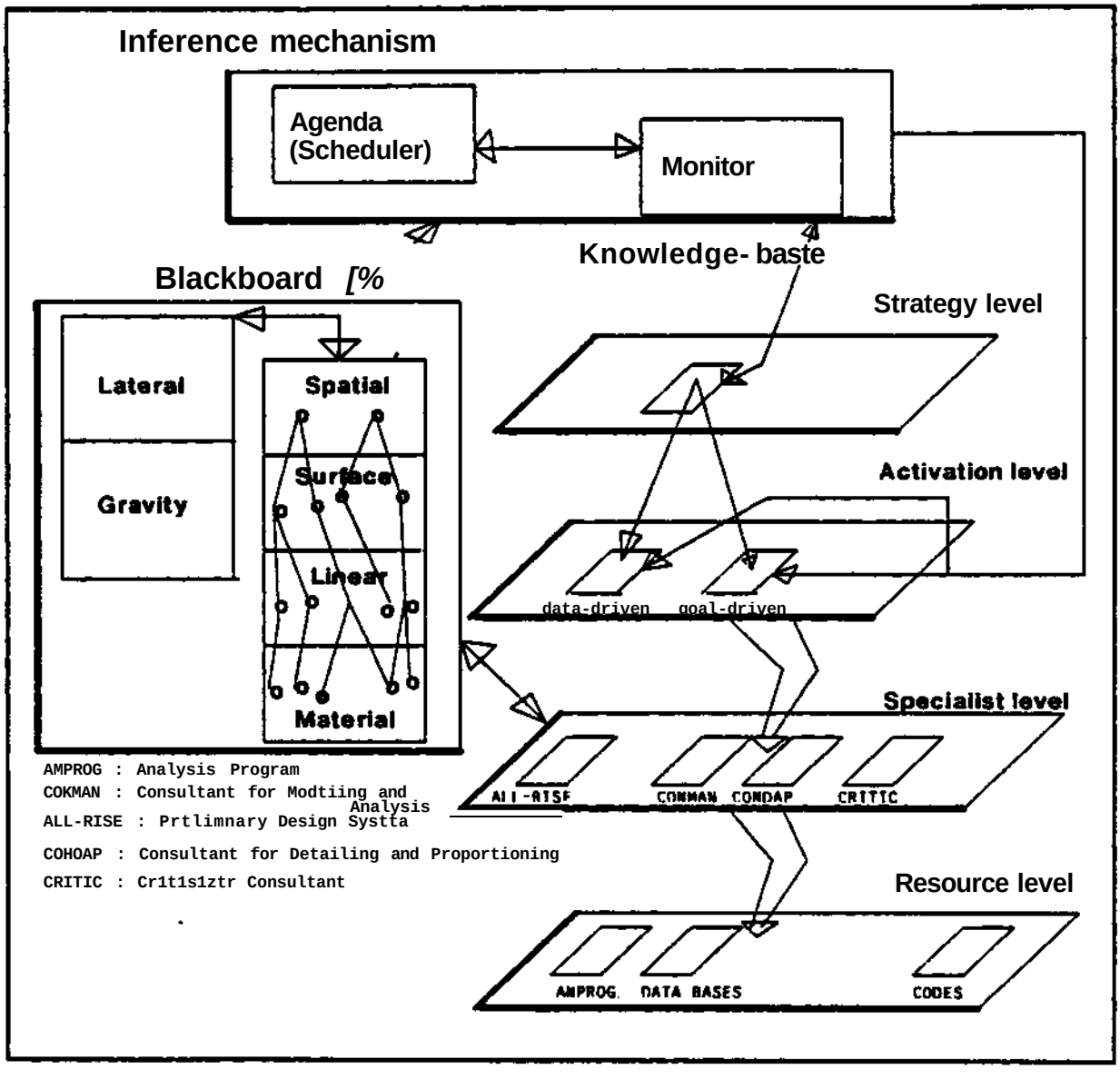



**Figure** 3: A Conceptual View of an Integrated Structural Analysis and Design System

HEARSAY-HI [3] or EMYCIN. However, their full-scale practical implementation will require higher-level expert system development tools, notably in the areas of user interfaces, explanation capabilities and knowledge-acquisition capabilities. Furthermore, practical implementations will also require close coupling to databases and algorithmic application programs.

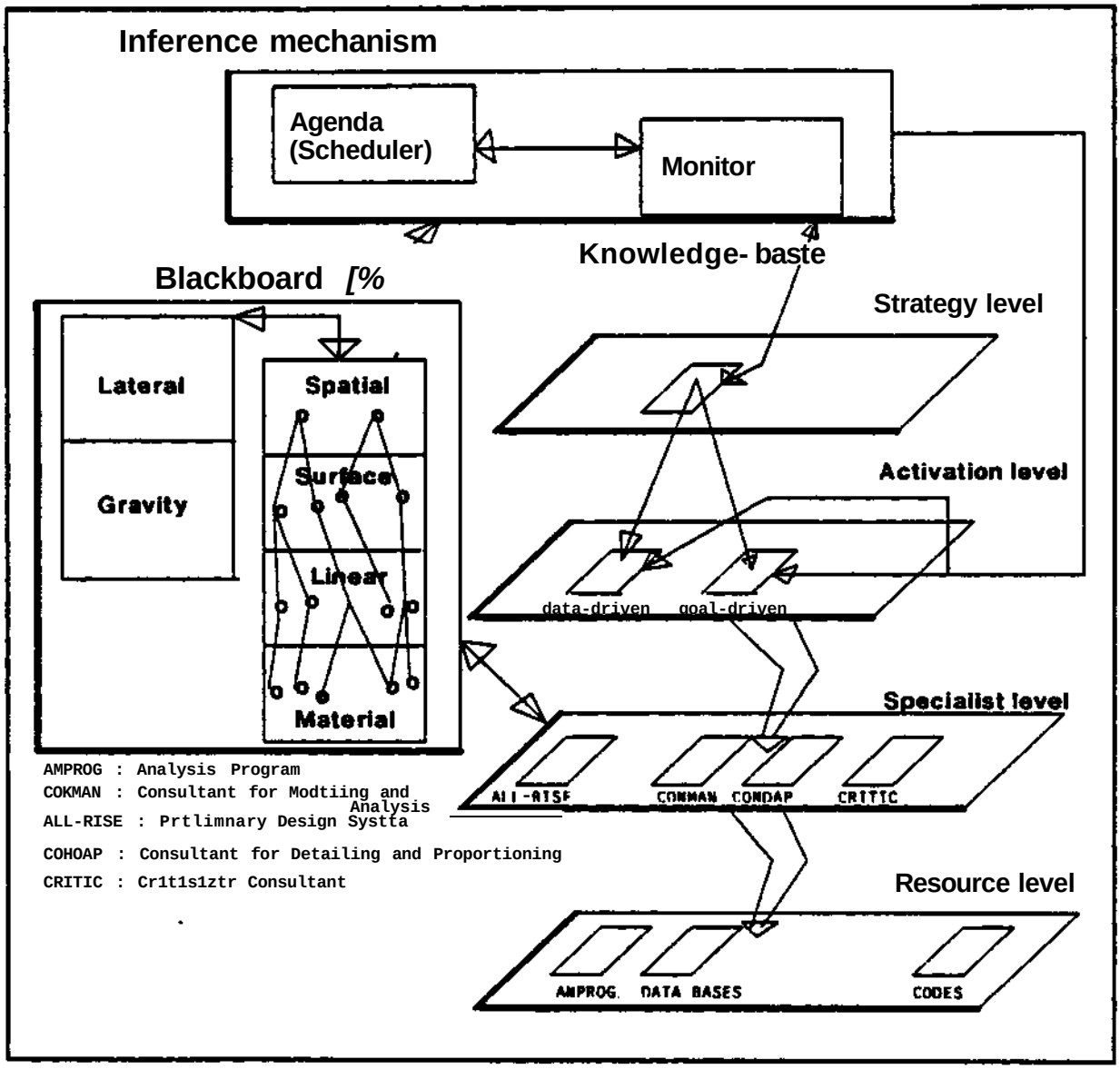



图3：集成式结构分析与设计系统HEARSAY-HI[3]或EMYCIN的概念视图。然而，其全面的工程实现将需要更高层次的专家系统开发工具，尤其是在用户界面、解释机制以及知识获取能力等方面。此外，实际应用还需要与数据库及算法型应用程序进行紧密耦合。

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